Real Analysis: Exercise 2

Due date:

Question 1. Find examples of two functions $f, g : \mathbb{R} \to \mathbb{R}$ such that $f \circ g$ and $g \circ f$ are not equal. Justify your answer.

Question 2. Let $f : X \to Y$ and $g : Y \to X$ be two functions such that

$$g \circ f(x) = x, \quad \forall\, x \in X.$$

Show that f is injective and g is surjective.

Question 3. Let $f : X \to Y$ and $g : Y \to Z$. Is it possible for $g \circ f$ to be a bijection if neither f or g are bijections? If not prove it, and if so give an example to justify it.

Question 4. Let $f : X \to Y$. Let C, D be subsets of X. Define

$$f(C) = \{f(x) : x \in C\}.$$

Prove that

$$f(C \cup D) = f(C) \cup f(D), \qquad f(C \cap D) \subset f(C) \cap f(D).$$

Give an example in which $f(C \cap D) \neq f(C) \cap f(D)$.

Question 5. Let $f : X \to Y$ be a function and let C be a subset of Y. Define the inverse image of C as

$$f^{-1}(C) := \{x \in X : f(x) \in C\}.$$

Prove that if C, D are subsets of Y, then

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D), \qquad f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D), \qquad f^{-1}(C^c) = \big(f^{-1}(C)\big)^c.$$

Here C^c denotes the complement.

Question 6.

1. Give an example that the intersection of infinitely many open sets is not an open set.
2. Give an example that the union of infinitely many closed sets is not a closed set.